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Power of one

How unilateral or **single-limb exercises** help athletes improve their stability and prevent injuries.

In the current landscape of modern physical preparation for team or individual sports, unilateral training has gained increasing attention from S&C coaches, athletic trainers, coaches and sports nutritionists. This approach, which focuses on performing exercises with a single limb at a time, contrasts with traditional bilateral training.

Unilateral training is characterised by exercises in which the load or resistance is applied to a single limb through its range of motion. This methodology aims to simulate specific motor actions that occur frequently in team sports, such as running, jumping, landing on one leg, changing direction, kicking and more.

Examples of unilateral exercises for the lower limbs include the Bulgarian squat or Rear Foot Elevated Split Squat (RFESS), Single-Leg Deadlift (SLDL), Single-Leg Countermovement Jump (SLCMJ), Single-Leg Drop Jump (SLDJ) and Single-Leg Broad Jump (SLBJ). In contrast, bilateral exercises involve both limbs simultaneously, such as the back squat, Countermovement Jump (CMJ), Drop Jump (DJ) and Standing Broad Jump (SBJ).

The distinction between these two approaches is not merely fashionable or a recent trend; it entails different neuromuscular challenges and potential physiological adaptations. By its very nature, unilateral training introduces greater variability, requiring increased activation of the stabilising muscles of the core and limbs. This facilitates the right movement execution while contributing to proprioception and motor control.

In many sports, unilateral actions are predominant. Running, for example, is a continuous sequence of single-leg supports, as well as many jumps and landings. Changes in direction — crucial in disciplines such as pace

bowling in cricket, football, basketball, hockey and handball — require the ability to generate and absorb force effectively on a single leg. In this context, a physical preparation programme that integrates unilateral training can provide a more direct transfer to sport-specific motions, enhancing the skills required on the field.

Potential benefits of integrating unilateral training

Reduction of Bilateral Deficit (BLD): BLD occurs when the maximum force produced by both limbs simultaneously is less than the sum of the maximum force produced by each limb individually. Research suggests that unilateral training can help reduce this deficit, improving recruitment capacity and intermuscular coordination during bilateral actions. Gonzalo-Skok et al. reported greater improvements in BLD among young basketball players who trained unilaterally compared with those who trained bilaterally.

Enhancement of Change Of Direction (COD) competence: Changes of direction are fundamental in many team sports. Studies have shown a strong